

MAE 3240 Homework 3

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Problem set number: #3

Due date: 02/24/2026

Acknowledgement:

No help

Problem #1:

Known: (only symbolically, no numerical values)

$$R, \dot{q}, T_{\infty}, h, T(x, t), k, \rho, c_p$$

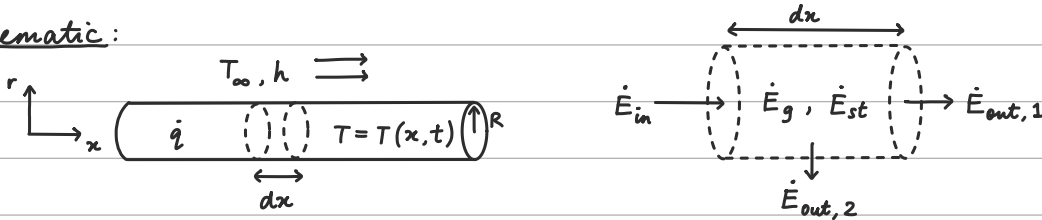
Find: a) Energy balance equation as fcn(T)

b) Governing differential equation

c) $T(x=0) = T_i$, $x=L$ exposed to air. At $t=0$, uniform $T = T_i$

→ boundary conditions and initial conditions

Schematic:



Properties: h, k, ρ, c_p

Assumptions: neglect thermal radiation

radius of the rod is v. small

temperature along radial direction is uniform → 1D

constant k, ρ, c_p

Analysis:

$$a) \dot{E}_{in} - \dot{E}_{out,1} - \dot{E}_{out,2} + \dot{E}_g = \dot{E}_{st}$$

$$q_x - q_{x+dx} - q_{conv} + \dot{q} = \dot{q}_{st}$$

$$\Rightarrow q_x - \left(q_x + \frac{dq_x}{dx} dx \right) - q_{conv} + \dot{q} = \dot{q}_{st}$$

$$\Rightarrow kA \frac{d^2 T}{dx^2} dx - hA_c (T - T_{\infty}) + \dot{q} A dx = \rho c_p A dx \frac{\partial T}{\partial t}$$

$$P = 2\pi R$$

$$A = \pi R^2$$

$$\Rightarrow \boxed{kA \frac{d^2 T}{dx^2} dx - hP dx (T - T_{\infty}) + \dot{q} A dx = \rho c_p A dx \frac{\partial T}{\partial t}}$$

$$b) \text{ Divide by } A dx \rightarrow k \frac{d^2 T}{dx^2} - \frac{hP}{A} (T - T_{\infty}) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\frac{P}{A} = \frac{2\pi R}{\pi R^2} = \frac{2}{R}$$

$$\Rightarrow \boxed{k \frac{d^2 T}{dx^2} - \frac{2h}{R} (T - T_{\infty}) = \rho c_p \frac{\partial T}{\partial t} - \dot{q}}$$

$$c) \boxed{\begin{array}{l} \text{BCs: } T(0, t) = T_i \\ \quad \quad \quad -k \frac{\partial T}{\partial x} \Big|_{x=L} = h(T(L, t) - T_{\infty}) \\ \text{IC: } T(x, 0) = T_i \text{ for } 0 \leq x \leq L \end{array}}$$

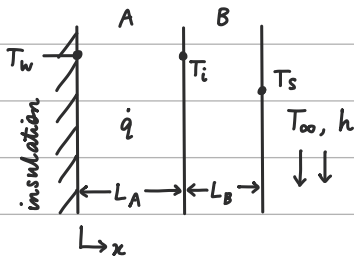
Problem #2:

Known: $L_A = 30 \text{ cm}$, $L_B = 10 \text{ cm}$, $k_B = 30 \text{ W/mK}$, $h = 500 \text{ W/m}^2\text{K}$

$$T_\infty = 20^\circ\text{C}, T_w = 115^\circ\text{C}, T_i = 100^\circ\text{C}$$

Find: a) T_s b) $\dot{q}$
c) k_A d) Temperature profile across walls A & B

Schematic:



Properties: $k_B = 30 \text{ W/mK}$, $h = 500 \text{ W/m}^2\text{K}$, $k_A = ?$

Assumptions: 1D, left wall thermally insulated,
no heat generation within B,
no contact thermal resistance b/w A & B

Analysis:

$$\begin{aligned} \text{a) } q''_{\text{cond},B} &= q''_{\text{conv}} \Rightarrow \frac{k_B}{L_B} (T_i - T_s) = h (T_s - T_\infty) \\ &\Rightarrow \frac{30}{0.1} (100 - T_s) = 500 (T_s - 20) \Rightarrow \boxed{T_s = 50^\circ\text{C}} \end{aligned}$$

$$\text{b) } q'' = \dot{q} L_A = \frac{k_B}{L_B} (T_i - T_s)$$

$$\Rightarrow \dot{q} (0.3) = \frac{30}{0.1} (100 - 50) = \boxed{50 \text{ kW/m}^3}$$

$$\text{c) } k \frac{d^2 T}{dx^2} + \dot{q} = 0 \rightarrow k \frac{dT}{dx} + \dot{q} x = c_1$$

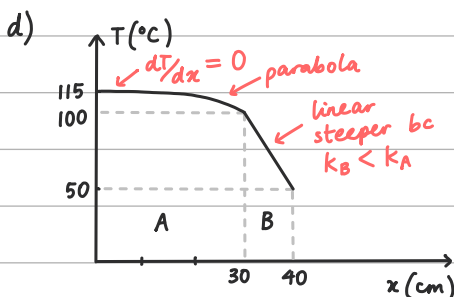
$$\left. \frac{dT}{dx} \right|_{x=0} = 0 \Rightarrow c_1 = 0 \rightarrow k T(x) + \dot{q} \frac{x^2}{2} = c_2$$

$$T|_{x=0} = T_w \Rightarrow k T_w = c_2$$

$$\Rightarrow T(x) = T_w - \frac{\dot{q} x^2}{2k}$$

$$T_i = T(L_A) = T_w - \frac{\dot{q} L_A^2}{2k_A} \Rightarrow 115 - \frac{50 \times 10^3 \times 0.3^2}{2k_A} = 100$$

$$\Rightarrow \boxed{k_A = 150 \text{ W/mK}}$$



Problem #3:

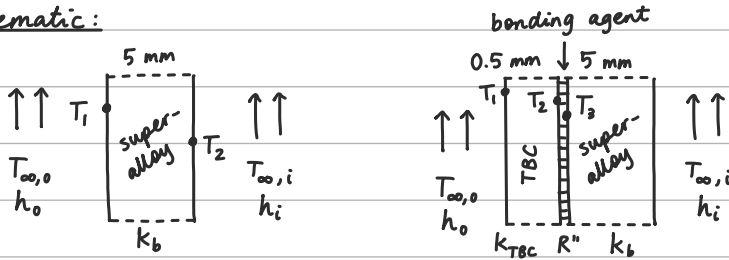
Known: $k_b = 25 \text{ W/mK}$, $k_{TBC} = 1 \text{ W/mK}$, $R'' = 10^{-4} \text{ m}^2 \cdot \text{K/W}$, $L_b = 5 \text{ mm}$, $L_{TBC} = 0.5 \text{ mm}$

$T_{\infty,o} = 1616 \text{ K}$, $T_{\infty,i} = 400 \text{ K}$, $T_{max} = 1200 \text{ K}$, $h_o = 1000 \text{ W/m}^2\text{K}$, $h_i = 500 \text{ W/m}^2\text{K}$

Find: a) Thermal resistance network w/ no TBC, $T_s < T_{max}$?

b) Thermal resistance network w/ TBC, $T_s < T_{max}$?

Schematic:



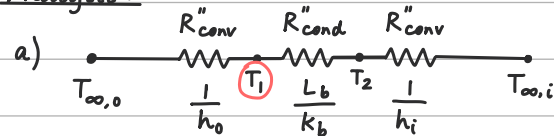
Properties: $k_b = 25 \text{ W/mK}$, $k_{TBC} = 1 \text{ W/mK}$, $h_o = 1000 \text{ W/m}^2\text{K}$, $h_i = 500 \text{ W/m}^2\text{K}$

Assumptions: metallic bonding agent has negligible thickness

steady state, neglect thermal radiation

blade is like plane wall $\rightarrow$ 1D conduction

Analysis:

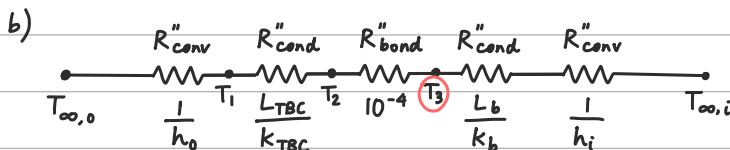


$$R''_{tot} = \frac{1}{1000} + \frac{5 \times 10^{-3}}{25} + \frac{1}{500} = 0.0032 \text{ m}^2\text{K/W}$$

$$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R''_{tot}} = \frac{1616 - 400}{0.0032} = 3.8 \times 10^5 \text{ W/m}^2$$

$$\Rightarrow 3.8 \times 10^5 = (1616 - T_1) 1000 \Rightarrow T_1 = 1236 \text{ K} > 1200 \text{ K} = T_{max}$$

superalloy blade cannot be maintained below T_{max}



$$R''_{tot} = \frac{1}{1000} + \frac{0.5 \times 10^{-3}}{1} + 10^{-4} + \frac{5 \times 10^{-3}}{25} + \frac{1}{500} = 0.0038 \text{ m}^2\text{K/W}$$

$$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R''_{tot}} = \frac{1616 - 400}{0.0038} = 3.2 \times 10^5 \text{ W/m}^2$$

$$\Rightarrow 3.2 \times 10^5 = (1616 - T_1) 1000 \Rightarrow T_1 = 1296 \text{ K}$$

$$3.2 \times 10^5 = (1296 - T_2) \frac{1}{0.5 \times 10^{-3}} \Rightarrow T_2 = 1136 \text{ K}$$

$$3.2 \times 10^5 = (1136 - T_3) \frac{1}{10^{-4}} \Rightarrow T_3 = 1104 \text{ K} < 1200 \text{ K} = T_{max}$$

superalloy blade can be maintained below T_{max}